Run Qafila locally

Setup, checks and supplier corrections

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1. Requirements

Use Python 3.10 through 3.13 and uv. Install uv using the instructions at docs.astral.sh/uv/getting-started/installation/. Open a new terminal after installation.

2. Get the project and environment

```
git clone https://github.com/Moashehri/qafila-margin-project.git
cd qafila-margin-project
uv sync --frozen
```

Run the remaining commands in this folder, beside pyproject.toml. On Windows, RUN_ON_WINDOWS.bat runs the setup, data generation, pipeline and both checkers in sequence.

3. Generate data and run the pipeline

```
uv run python tools/make_data.py
uv run python run.py
```

For seed 2026, expect 633 cost versions, 165 weekly margin rows and 7 quarantined rows before the correction. The dbt build contains 8 models and 26 tests; all 34 build nodes should pass.

4. Check the results

```
uv run python tools/check.py
uv run python tools/reference_check.py
uv run python -m unittest discover -s tests -v
```

Expected: 22/22 contract checks, zero reference differences, and 3 passing regression tests. The reference checker recomputes the results directly from the original inputs.

5. Inspect the output

```
uv run python tools/show_tables.py
uv run python tools/spot_check.py
```

Re-runs and corrections

6. Re-run unchanged inputs

```
uv run python run.py
```

Each published table should report unchanged and no new snapshot. The second run must not duplicate source rows or change table contents.

7. Add the late supplier correction

```
uv run python tools/make_data.py --add-correction
uv run python run.py
uv run python tools/reference_check.py
uv run python tools/show_tables.py
```

Cost history increases to 637 versions on seed 2026. The weekly margin table receives one overwrite snapshot. The quarantine stays at 7 rows, with its original snapshot. The table display also reads historical snapshots to show totals from before the correction.

8. Run the complete verification

```
uv run python tools/verify_pipeline.py
# Optional machine-readable report:
uv run python tools/verify_pipeline.py --report tmp/results.json
```

This creates temporary datasets for seeds 2026 and 11, checks repeated runs and corrections, reads older snapshots, and removes its temporary data. It does not change the default data or warehouse.

Common problems

Message	Action
uv not recognized	Install uv and reopen the terminal.
No pyproject.toml found	Change into the cloned project folder.
qafila_oltp.sqlite not found	Run tools/make_data.py first.
dbt build failed	Read the failing model or test in the output.
DuckDB file locked	Close any other connection to the same warehouse.

Keep the warehouse to retain historical snapshots. Datasets, warehouses and build outputs are ignored by Git. See README.md for alternate dataset paths and the full design.